

Enterprise Knowledge Graph & Relational Operations Audit

Overview: This technical audit details the organizational taxonomy, inter-entity correlation matrices, database transactional schemas, and departmental budget pivot analyses for enterprise graph construction.

Figure 2: Knowledge Graph Node-Link Topology

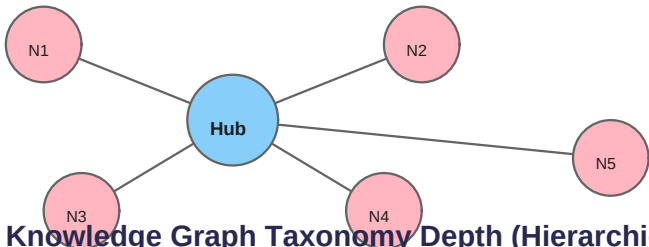


Table 2.1: Knowledge Graph Taxonomy Depth (Hierarchical)

Ontology Branch	Sub-Classes	Sub-Classes	Properties	Properties
	Count	Depth	Required	Optional
Entity.Organization	14	3	8	24
Entity.FinancialInstrument	22	4	15	45
Event.CorporateAction	8	2	6	18

Table 2.2: Semantic Relationship Co-occurrence Matrix (Matrix)

	Entity.Org	Entity.Person	Entity.Security	Event.Trade
Entity.Org	1.00	0.85	0.92	0.64
Entity.Person	0.85	1.00	0.41	0.78
Entity.Security	0.92	0.41	1.00	0.95
Event.Trade	0.64	0.78	0.95	1.00

Database Linkage Schema: Tabular entities are cross-referenced across internal relational stores using explicit foreign key mappings and transaction logs.

Table 2.3: Entity Transaction Ledger with Foreign Keys (Relational)

tx_id	entity_fk	graph_node_fk	tx_type	amount_usd
TX-9901	ORG-771	NODE-0012	ACQUISITION	1,200,000,000
TX-9902	ORG-882	NODE-0045	DIVIDEND	45,000,000
TX-9903	ORG-771	NODE-0089	STOCK_BUYBACK	250,000,000

Table 2.4: Cross-Sector Budget Allocation Pivot (Pivot / Cross-tab)

Corporate Sector	2023 Actual	2023 Actual	2024 Budget	2024 Budget
	Capex (\$M)	Opex (\$M)	Capex (\$M)	Opex (\$M)
Information Technology	450.0	120.5	580.0	145.0
Financial Services	210.0	340.0	260.0	390.0
Healthcare Systems	180.0	95.0	220.0	110.0